



Advanced Cisco Networking Academy: Designing a Multi-Area OSPF Network

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Purpose

The configuration of multi-area OSPF teaches core concepts on how routing works and how to effectively segment a network into different areas to simplify the network's topology. The configuration of OSPF additionally teaches how dynamic routing protocols work, allowing an understanding of other routing protocols besides OSPF.

Background

OSPF or Open Shortest Path First is a dynamic networking protocol that allows for easier more efficient route configuration on routers. Dynamic routing protocols are a vital part of enterprise networking as it allows for network redundancy another vital and foundational part of network security that enables company operation to continue even in the event that a network device goes down for any reason ensuring that operations can continue without interruptions. Additionally, OSPF allows for network engineers and administrators to save time eliminating the need to enter every route to every network into each router allowing for more time to be spent on more urgent and pressing tasks, this also applies to when a network device goes down as without a dynamic routing protocol like OSPF a network administrator/engineer would need to identify which device went down and find an alternative route for company traffic to access necessary resources until the administrator identifies the problem, but this problem can be avoided with dynamic routing protocols like OSPF which provide that redundancy needed to keep the network alive giving the administrator more time to find out what happened to the device.

OSPF provides these advantages by building a routing table that is made by gathering link state information to build a topology of the network. OSPF analyses all links through the network topology it built, link metrics (like link bandwidth), and Dijkstra's algorithm to calculate the most efficient path to send network traffic to provide the fastest connection to its users. OSPF can be run with multiple areas to simplify the network, all areas must be connected to the backbone area and when running single-area OSPF you must use area 0 the backbone area. When running multi-area OSPF it's necessary to configure a ABR or an Area Border Router which lies on the border of the backbone and the area connected to it, this router is needed due to the fact that OSPF enabled routers can only build a topology of the area it is in creating the need to connect the two areas thereby the creation of the ABR. Additionally, OSPF can be run with both IPv4 and IPv6, OSPF can be configured to use internal authentication to only allow trusted routers to participate in routing when using IPv4, but IPv6 implementations only rely on the standard IPsec protocols for authentication.

For OSPF enabled devices to build a routing table that summarizes the whole area, two devices need to exchange link-state information. For two OSPF routers to exchange this information they need to form adjacencies and to form these adjacencies the

router must go through the neighbor state exchanges which consist of the exchange of hello packets. Hello packets are packets which an OSPF enabled router sends out to active interfaces to check for any adjacent routers to form adjacencies. After each router has sent their hello packets, they can start the database exchange where each router sends its link state information to each other allowing for an updated topology table. Whenever another OSPF router is added to an area each routers topology will update through this process but instead of starting from the neighbor state exchanges they will send updated LSA's or Link state Advertisements with the updated topology as the routers will have already established adjacencies. Because an OSPF router will send hello packets to all interfaces if your router is sending advertisements to a group of users, they could send malicious hello packets back to form an adjacency to disrupt the company network. To prevent this, you can set interfaces pointing towards users to passive interfaces which still advertise the network but does not send hello packets to ensure security.

When many OSPF routers share one link it can complicate the network, slow it down, and make it hard to manage. To prevent this complication the routers, go through an election process to simplify the network, the winner of this election is determined by the set priority value that ranges from zero meaning it can never be the DR or the winner and 255 meaning it will always win, if all priority values are non-zero and the same the router with the highest IP address will become the DR this IP address can either be from a physical interface or a manually set one. Once a router is elected DR the BDR is elected which is the router with the second highest Priority or IP address. The DR and BDR each form one connection to the non-elected routers instead of each router having a connection to each other simplifying the network. These non-elected routers are called DROTHER and are stuck in the 2-way exchange state with only adjacency to the DR and BDR.

Lab Summary

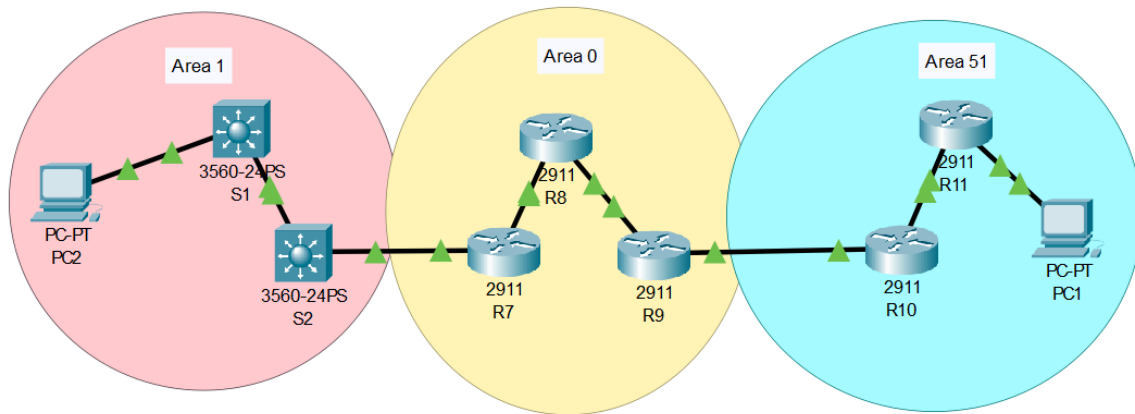
In this lab there are seven OSPF-enabled devices with 3 different areas including the backbone area. There are 2 devices in each non-backbone area with one area consisting of two layer 3 switches and the other 5 devices being routers. The layer 3 switches are put into area 1, 3 routers into area 0, 2 routers into area 51, and the border routers are placed so that there are 3 routers completely in area 0. Each active interface was advertised in its respective area, and each router was given a router id based on its hostname. All devices operated in process 1.

Lab Commands

- Router ospf [1-65535] – enables OSPF on the router and puts the user into the OSPF router

- Router-id x.x.x.x - configures a router id for the OSPF router, these IDs have to be unique over an area
- Ip ospf x area x – configures the interface to advertise the network the interface is on to other OSPF routers
- Ip routing – enables routing on the router and switch
- No switchport – makes the switchport on the layer 3 switch able to route at the layer 3 level

Network Diagram



Switch 1 (Area 1)	G1/0/2 (PC1)	G1/0/1 (S2 G1/0/1)
	192.168.1.1	192.168.2.1
Switch 2 (Area 1)	G1/0/2 (R7 g0/0)	G1/0/1 (S1 G1/0/1)
	192.168.3.1	192.168.2.2
Router 7 (Area 0)	G0/0/0 (S2 G1/0/2)	G0/0/1 (R8 G0/0/0)
	192.168.3.2	192.168.4.1
Router 8 (Area 0)	G0/0/0 (R7 G0/0/1)	G0/0/1 (R9 G0/0/0)
	192.168.4.2	192.168.5.1
Router 9 (Area 0)	G0/0/0 (R8 G0/0/1)	G0/0/1 (R10 G0/0/0)
	192.168.5.2	192.168.6.1
Router 10 (Area 51)	G0/0/0 (R9 G0/0/1)	G0/0/1 (R11 G0/0/0)
	192.168.6.2	192.168.7.1
Router 11 (Area 51)	G0/0/0 (R10 G0/0/1)	G0/0/1 (PC2)
	192.168.7.2	192.168.8.1

PC1 (Area 1)	192.168.1.2
PC2 (Area 51)	192.168.8.2

Configurations

Switch 1

S1#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

Gateway of last resort is not set

```
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet1/0/2
L    192.168.1.1/32 is directly connected, GigabitEthernet1/0/2
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/30 is directly connected, GigabitEthernet1/0/1
L    192.168.2.1/32 is directly connected, GigabitEthernet1/0/1
192.168.3.0/30 is subnetted, 1 subnets
O IA  192.168.3.0 [110/2] via 192.168.2.2, 00:53:43, GigabitEthernet1/0/1
192.168.4.0/30 is subnetted, 1 subnets
O IA  192.168.4.0 [110/3] via 192.168.2.2, 00:49:23, GigabitEthernet1/0/1
192.168.5.0/30 is subnetted, 1 subnets
O IA  192.168.5.0 [110/4] via 192.168.2.2, 00:48:39, GigabitEthernet1/0/1
192.168.6.0/30 is subnetted, 1 subnets
O IA  192.168.6.0 [110/5] via 192.168.2.2, 00:45:52, GigabitEthernet1/0/1
192.168.7.0/30 is subnetted, 1 subnets
O IA  192.168.7.0 [110/6] via 192.168.2.2, 00:44:04, GigabitEthernet1/0/1
O IA  192.168.8.0/24 [110/7] via 192.168.2.2, 00:43:29, GigabitEthernet1/0/1
```

S1#show ipv6 route

IPv6 Routing Table - default - 11 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
R - RIP, D - EIGRP, EX - EIGRP external, ND - ND Default
NDp - ND Prefix, DCE - Destination, NDr - Redirect, O - OSPF Intra
OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
ON2 - OSPF NSSA ext 2
C 2001:DB8:ACAD:1::/64 [0/0]
via GigabitEthernet1/0/2, directly connected
L 2001:DB8:ACAD:1::1/128 [0/0]

```

    via GigabitEthernet1/0/2, receive
C  2001:DB8:ACAD:2::/64 [0/0]
    via GigabitEthernet1/0/1, directly connected
L  2001:DB8:ACAD:2::1/128 [0/0]
    via GigabitEthernet1/0/1, receive
OI 2001:DB8:ACAD:3::/64 [110/2]
    via FE80::B28B:D0FF:FED6:C64, GigabitEthernet1/0/1
OI 2001:DB8:ACAD:4::/64 [110/3]
    via FE80::B28B:D0FF:FED6:C64, GigabitEthernet1/0/1
OI 2001:DB8:ACAD:5::/64 [110/4]
    via FE80::B28B:D0FF:FED6:C64, GigabitEthernet1/0/1
OI 2001:DB8:ACAD:6::/64 [110/5]
    via FE80::B28B:D0FF:FED6:C64, GigabitEthernet1/0/1
OI 2001:DB8:ACAD:7::/64 [110/6]
    via FE80::B28B:D0FF:FED6:C64, GigabitEthernet1/0/1
OI 2001:DB8:ACAD:8::/64 [110/7]
    via FE80::B28B:D0FF:FED6:C64, GigabitEthernet1/0/1
L  FF00::/8 [0/0]
    via Null0, receive

```

S1#show ip ospf neigh detail

```

Neighbor 2.2.2.2, interface address 192.168.2.2, interface-id 39
  In the area 51 via interface GigabitEthernet1/0/1
  Neighbor priority is 1, State is FULL, 6 state changes
  DR is 192.168.2.1 BDR is 192.168.2.2
  Options is 0x12 in Hello (E-bit, L-bit)
  Options is 0x52 in DBD (E-bit, L-bit, O-bit)
  LLS Options is 0x1 (LR)
  Dead timer due in 00:00:35
  Neighbor is up for 23:57:39
  Index 1/1/1, retransmission queue length 0, number of retransmission 0
  First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
  Last retransmission scan length is 0, maximum is 0
  Last retransmission scan time is 0 msec, maximum is 0 msec

```

S1#show ip ospf int brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi1/0/1	1	51	192.168.2.1/30	1	DR	1/1	
Gi1/0/2	1	51	192.168.1.1/24	1	DR	0/0	

S1#show ipv6 neigh detail

IPv6 Address	TRLV	Age	Link-layer Addr	State	Interface
2001:DB8:ACAD:1::2	0	28	04d9.c8ba.2574	STALE	Gi1/0/2
2001:DB8:ACAD:1:705D:3E65:F064:1A11	0	0	04d9.c8ba.2574	REACH	Gi1/0/2
FE80::19A5:5D31:E6D0:6DA	0	0	04d9.c8ba.2574	REACH	Gi1/0/2
FE80::B28B:D0FF:FED6:C64	0	36	b08b.d0d6.0c64	STALE	Gi1/0/1

Switch 2

S2#show run

Building configuration...

Current configuration : 9345 bytes

!

! Last configuration change at 14:56:49 UTC Wed Sep 24 2025

!

version 17.3

service timestamps debug datetime msec

service timestamps log datetime msec

! Call-home is enabled by Smart-Licensing.

service call-home

platform punt-keepalive disable-kernel-core

!

hostname S2

!

!

vrf definition Mgmt-vrf

!

address-family ipv4

exit-address-family

!

address-family ipv6

exit-address-family

!

!

!

!

!

no aaa new-model

switch 1 provision c9200l-24t-4g

!

!

!

!

!

!

!

!

!

!

!

ip routing

!

!

!
!
login on-success log
ipv6 unicast-routing
no device-tracking logging theft
!
crypto pki trustpoint TP-self-signed-2577419285
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-2577419285
revocation-check none
rsakeypair TP-self-signed-2577419285
!
crypto pki trustpoint SLA-TrustPoint
enrollment pkcs12
revocation-check crl
!
!
crypto pki certificate chain TP-self-signed-2577419285
certificate self-signed 01
30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 32353737 34313932 3835301E 170D3235 30393234 31343433
31335A17 0D333530 39323431 34343331 335A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32 35373734
31393238 35308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201 0100BF8F C6D325BF 986B201D 487C7B73 0F16CFB0 8369A1C0 34E91FCF
3C18A99A 3D755EC4 6C2E66B5 5C3A00D4 FC980261 15E7D936 07481731
FD76A7B5
744D8EA9 D561AC0B D6F115E7 61CD2A17 D91E61D2 FF8C00FD 73F213BD
2974F619
B175D974 1FCCB579 ECA51034 B16B55F2 CC91BACA 30B6DC9D 0E25A951
95F18224
AF6BA461 EE02A0BB 81DCBC42 E4BCECF5 5D7FD12C DB2D7383 8506483E
A4362F2B
73D526FA 25D89C93 A851DD93 532FDDA9 42E6602E C03044E7 8A21D675
ADF95A18
6C7C18C9 F815420D A481D4DA 2D23FB7D 6BB46BAF 6D740A19 28216F58
EE8D9136
F26D58BD 66C63282 A181DA32 24BD2188 679B6BA5 3BE01A5D 6DE9D713
8BB04339
CF573E43 971B0203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603 551D2304 18301680 14DCF50A B4F70B82 D2B1FFDD E610E46A
960B379A
7C301D06 03551D0E 04160414 DCF50AB4 F70B82D2 B1FFDDE6 10E46A96

0B379A7C

300D0609 2A864886 F70D0101 05050003 82010100 A107032D 5A0C16E4 83F962ED
C66C600F CD7D43E7 EA14DB50 3EE9B8B9 7B2EA40A 005AE896 BDD75BA0

63953D05

3D63F65D 4E5AD350 D79DC07B 3B424527 91A6A28C CD2B519C 8DCCA9C4

36699103

E80A8E4F 2FAE71D2 25F41F03 70DD7069 51F7E47F A75C743D 8EB96B73

F4AAB96C

5ACD96F4 0444F453 283FD57F 99863177 22B8F884 A4C4B588 5E07543B 92FF92FE
9FBA131A BE21EF6F 27594921 FB98D6A5 44073ECA 2D002E9F 7F10C9A0

E0120DC2

CDC7980B 55E73C0B 41112F28 B7F52B24 8EDFF1CF E11BF2DA 23D4055A

1045C740

8DF787EC F015A4AF 2B54CBA9 B0570C1C 26D351D7 37EB135F E204DD27

2D13C165

94B6A16A E4C99AC3 377D02E3 CDEA2D43 AD84FCBC

quit

crypto pki certificate chain SLA-TrustPoint

certificate ca 01

30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101 0B050030
32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317 43697363
6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533 30313934
3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355 040A1305
43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73 696E6720
526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1

F1EFF64D

CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388

8A38E520

1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7

D8F256EE

4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F

EA2956AC

7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF

58BD7188

68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B

42C68BB7

C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8

8F27D191

C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368

95135E44

DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04 04030201
06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604 1449DC85
4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01 010B0500

03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78 240DA905
604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1
6C9E3D8B
D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146
8DFC66A8
467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2
55A9232C
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49
1765308B
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69 39F08678
80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD
230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F 719BB2F0
D697DF7F 28

--More--

!

license boot level network-essentials addon dna-essentials

!

!

diagnostic bootup level minimal

!

spanning-tree mode rapid-pvst

spanning-tree extend system-id

memory free low-watermark processor 10308

!

!

redundancy

mode sso

!

!

transceiver type all

monitoring

!

!

class-map match-any system-cpp-police-ewlc-control

description EWLC Control

class-map match-any system-cpp-police-topology-control

description Topology control

class-map match-any system-cpp-police-sw-forward

description Sw forwarding, L2 LVX data packets, LOGGING, Transit Traffic

class-map match-any system-cpp-default

description EWLC data, Inter FED Traffic

class-map match-any system-cpp-police-sys-data

description Openflow, Exception, EGR Exception, NFL Sampled Data, RPF Failed

```

class-map match-any system-cpp-police-punt-webauth
  description Punt Webauth
class-map match-any system-cpp-police-l2lvx-control
  description L2 LVX control packets
class-map match-any system-cpp-police-forus
  description Forus Address resolution and Forus traffic
class-map match-any system-cpp-police-multicast-end-station
  description MCAST END STATION
class-map match-any system-cpp-police-high-rate-app
  description High Rate Applications
class-map match-any system-cpp-police-multicast
  description MCAST Data
class-map match-any system-cpp-police-l2-control
  description L2 control
class-map match-any system-cpp-police-dot1x-auth
  description DOT1X Auth
class-map match-any system-cpp-police-data
  description ICMP redirect, ICMP_GEN and BROADCAST
class-map match-any system-cpp-police-stackwise-virt-control
  description Stackwise Virtual OOB
class-map match-any non-client-nrt-class
class-map match-any system-cpp-police-routing-control
  description Routing control and Low Latency
class-map match-any system-cpp-police-protocol-snooping
  description Protocol snooping
class-map match-any system-cpp-police-dhcp-snooping
  description DHCP snooping
class-map match-any system-cpp-police-ios-routing
  description L2 control, Topology control, Routing control, Low Latency
class-map match-any system-cpp-police-system-critical
  description System Critical and Gold Pkt
class-map match-any system-cpp-police-ios-feature
  description
ICMPGEN,BROADCAST,ICMP,L2LVXCntrl,ProtoSnoop,PuntWebauth,MCASTData,Tra
nsit,DOT1XAuth,Swfwd,LOGGING,L2LVXData,ForusTraffic,ForusARP,McastEndStn,O
penflow,Exception,EGRExcption,NflSampled,RpfFailed
!
policy-map system-cpp-policy
!
!
!
!
!
!
!

```

```
!  
!  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
vrf forwarding Mgmt-vrf  
no ip address  
shutdown  
negotiation auto  
!  
interface GigabitEthernet1/0/1  
no switchport  
ip address 192.168.2.2 255.255.255.252  
ip ospf 1 area 1  
ipv6 address 2001:DB8:ACAD:2::2/64  
ipv6 ospf 1 area 1  
!  
interface GigabitEthernet1/0/2  
no switchport  
ip address 192.168.3.1 255.255.255.252  
ip ospf 1 area 0  
ipv6 address 2001:DB8:ACAD:3::1/64  
ipv6 ospf 1 area 0  
!  
interface GigabitEthernet1/0/3  
!  
interface GigabitEthernet1/0/4  
!  
interface GigabitEthernet1/0/5  
!  
interface GigabitEthernet1/0/6  
!  
interface GigabitEthernet1/0/7  
!  
interface GigabitEthernet1/0/8  
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!
```

```
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
router-id 2.2.2.2
!
ip forward-protocol nd
```

```

ip http server
ip http authentication local
ip http secure-server
!
!
ipv6 router ospf 1
router-id 2.2.2.2
!
!
!
control-plane
service-policy input system-cpp-policy
!
!
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
transport input ssh
line vty 5 15
login
transport input ssh
!
call-home
! If contact email address in call-home is configured as sch-smart-
licensing@cisco.com
! the email address configured in Cisco Smart License Portal will be used as contact
email address to send SCH notifications.
contact-email-addr sch-smart-licensing@cisco.com
profile "CiscoTAC-1"
  active
  destination transport-method http
!
!
!
!
!
!
end
S2(config-if)#do show ip ospf database

```

OSPF Router with ID (2.2.2.2) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
2.2.2.2	2.2.2.2	175	0x8000003B	0x00F616	1
7.7.7.7	7.7.7.7	1917	0x80000006	0x00E447	2
8.8.8.8	8.8.8.8	15	0x80000005	0x00EA36	2
9.9.9.9	9.9.9.9	1686	0x80000005	0x00CC49	2
10.10.10.10	10.10.10.10	1655	0x80000003	0x0055A8	1

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.3.1	2.2.2.2	175	0x80000002	0x00168C
192.168.4.2	8.8.8.8	15	0x80000002	0x001957
192.168.5.1	8.8.8.8	15	0x80000002	0x007CEB
192.168.6.1	9.9.9.9	1686	0x80000001	0x00A9B2

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.1.0	2.2.2.2	417	0x80000002	0x00A225
192.168.2.0	2.2.2.2	417	0x80000002	0x007B4F
192.168.7.0	10.10.10.10	1645	0x80000001	0x005551
192.168.8.0	10.10.10.10	1610	0x80000001	0x00663B

Router Link States (Area 51)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	458	0x80000049	0x002C62	2
2.2.2.2	2.2.2.2	417	0x80000034	0x00FC18	1

Net Link States (Area 51)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.2.1	1.1.1.1	708	0x8000002B	0x00CFC6

Summary Net Link States (Area 51)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.3.0	2.2.2.2	417	0x80000002	0x007059
192.168.4.0	2.2.2.2	1961	0x80000001	0x007157
192.168.5.0	2.2.2.2	1916	0x80000001	0x007056
192.168.6.0	2.2.2.2	1749	0x80000001	0x006F55


```
192.168.7.0 2.2.2.2 1641 0x80000001 0x006E54
192.168.8.0 2.2.2.2 1606 0x80000001 0x007F3E
```

S2(config-if)#do show ip ospf neigh detail

Neighbor 7.7.7.7, interface address 192.168.3.2

In the area 0 via interface GigabitEthernet1/0/2

Neighbor priority is 1, State is FULL, 6 state changes

DR is 192.168.3.1 BDR is 192.168.3.2

Options is 0x12 in Hello (E-bit, L-bit)

Options is 0x52 in DBD (E-bit, L-bit, O-bit)

LLS Options is 0x1 (LR)

Dead timer due in 00:00:36

Neighbor is up for 00:21:32

Index 1/1/2, retransmission queue length 0, number of retransmission 0

First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0

Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 1.1.1.1, interface address 192.168.2.1, interface-id 39

In the area 51 via interface GigabitEthernet1/0/1

Neighbor priority is 1, State is FULL, 6 state changes

DR is 192.168.2.1 BDR is 192.168.2.2

Options is 0x12 in Hello (E-bit, L-bit)

Options is 0x52 in DBD (E-bit, L-bit, O-bit)

LLS Options is 0x1 (LR)

Dead timer due in 00:00:36

Neighbor is up for 23:24:50

Index 1/1/1, retransmission queue length 0, number of retransmission 0

First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0

Last retransmission scan time is 0 msec, maximum is 0 msec

S2(config-if)#do show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP

n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

H - NHRP, G - NHRP registered, g - NHRP registration summary

o - ODR, P - periodic downloaded static route, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O 192.168.1.0/24 [110/2] via 192.168.2.1, 00:35:06, GigabitEthernet1/0/1
 192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
 C 192.168.2.0/30 is directly connected, GigabitEthernet1/0/1
 L 192.168.2.2/32 is directly connected, GigabitEthernet1/0/1
 192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
 C 192.168.3.0/30 is directly connected, GigabitEthernet1/0/2
 L 192.168.3.1/32 is directly connected, GigabitEthernet1/0/2
 192.168.4.0/30 is subnetted, 1 subnets
 O 192.168.4.0 [110/2] via 192.168.3.2, 00:30:02, GigabitEthernet1/0/2
 192.168.5.0/30 is subnetted, 1 subnets
 O 192.168.5.0 [110/3] via 192.168.3.2, 00:30:02, GigabitEthernet1/0/2
 192.168.6.0/30 is subnetted, 1 subnets
 O 192.168.6.0 [110/4] via 192.168.3.2, 00:27:15, GigabitEthernet1/0/2
 192.168.7.0/30 is subnetted, 1 subnets
 O IA 192.168.7.0 [110/5] via 192.168.3.2, 00:25:27, GigabitEthernet1/0/2
 O IA 192.168.8.0/24 [110/6] via 192.168.3.2, 00:24:52, GigabitEthernet1/0/2

S2(config-if)#do show ipv6 ospf neigh detail

OSPFv3 Router with ID (2.2.2.2) (Process ID 1)

Neighbor 7.7.7.7

In the area 0 via interface GigabitEthernet1/0/2
 Neighbor: interface-id 6, link-local address FE80::B6A8:B9FF:FE01:B510
 Neighbor priority is 1, State is FULL, 6 state changes
 DR is 7.7.7.7 BDR is 2.2.2.2
 Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
 Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
 Dead timer due in 00:00:35
 Neighbor is up for 00:14:38
 Index 1/1/2, retransmission queue length 0, number of retransmission 0
 First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
 Last retransmission scan length is 0, maximum is 0
 Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 1.1.1.1

In the area 51 via interface GigabitEthernet1/0/1
 Neighbor: interface-id 39, link-local address FE80::BED2:95FF:FE3C:4AE4
 Neighbor priority is 1, State is FULL, 6 state changes
 DR is 1.1.1.1 BDR is 2.2.2.2
 Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
 Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
 Dead timer due in 00:00:36
 Neighbor is up for 00:15:05

Index 1/1/1, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

S2#show ipv6 route

IPv6 Routing Table - default - 11 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

R - RIP, D - EIGRP, EX - EIGRP external, ND - ND Default

NDp - ND Prefix, DCE - Destination, NDr - Redirect, O - OSPF Intra

OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1

ON2 - OSPF NSSA ext 2

O 2001:DB8:ACAD:1::/64 [110/2]
via FE80::BED2:95FF:FE3C:4AE4, GigabitEthernet1/0/1
C 2001:DB8:ACAD:2::/64 [0/0]
via GigabitEthernet1/0/1, directly connected
L 2001:DB8:ACAD:2::2/128 [0/0]
via GigabitEthernet1/0/1, receive
C 2001:DB8:ACAD:3::/64 [0/0]
via GigabitEthernet1/0/2, directly connected
L 2001:DB8:ACAD:3::1/128 [0/0]
via GigabitEthernet1/0/2, receive
O 2001:DB8:ACAD:4::/64 [110/2]
via FE80::B6A8:B9FF:FE01:B510, GigabitEthernet1/0/2
O 2001:DB8:ACAD:5::/64 [110/3]
via FE80::B6A8:B9FF:FE01:B510, GigabitEthernet1/0/2
O 2001:DB8:ACAD:6::/64 [110/4]
via FE80::B6A8:B9FF:FE01:B510, GigabitEthernet1/0/2
OI 2001:DB8:ACAD:7::/64 [110/5]
via FE80::B6A8:B9FF:FE01:B510, GigabitEthernet1/0/2
OI 2001:DB8:ACAD:8::/64 [110/6]
via FE80::B6A8:B9FF:FE01:B510, GigabitEthernet1/0/2
L FF00::/8 [0/0]
via Null0, receive

S2#show ip ospf int brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi1/0/2	1	0	192.168.3.1/30	1	DR	1/1	
Gi1/0/1	1	51	192.168.2.2/30	1	BDR	1/1	

Router 7

R7#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O IA 192.168.1.0/24 [110/3] via 192.168.3.1, 00:55:31, GigabitEthernet0/0/0
192.168.2.0/30 is subnetted, 1 subnets
O IA 192.168.2.0 [110/2] via 192.168.3.1, 00:55:31, GigabitEthernet0/0/0
192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.3.0/30 is directly connected, GigabitEthernet0/0/0
L 192.168.3.2/32 is directly connected, GigabitEthernet0/0/0
192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.4.0/30 is directly connected, GigabitEthernet0/0/1
L 192.168.4.1/32 is directly connected, GigabitEthernet0/0/1
192.168.5.0/30 is subnetted, 1 subnets
O 192.168.5.0 [110/2] via 192.168.4.2, 00:52:09, GigabitEthernet0/0/1
192.168.6.0/30 is subnetted, 1 subnets
O 192.168.6.0 [110/3] via 192.168.4.2, 00:49:21, GigabitEthernet0/0/1
192.168.7.0/30 is subnetted, 1 subnets
O IA 192.168.7.0 [110/4] via 192.168.4.2, 00:47:38, GigabitEthernet0/0/1
O IA 192.168.8.0/24 [110/5] via 192.168.4.2, 00:47:03, GigabitEthernet0/0/1

R7#show ipv6 route

IPv6 Routing Table - default - 11 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

OI 2001:DB8:ACAD:1::/64 [110/3]
via FE80::B28B:D0FF:FED6:C56, GigabitEthernet0/0/0
OI 2001:DB8:ACAD:2::/64 [110/2]
via FE80::B28B:D0FF:FED6:C56, GigabitEthernet0/0/0
C 2001:DB8:ACAD:3::/64 [0/0]
via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:3::2/128 [0/0]
via GigabitEthernet0/0/0, receive
C 2001:DB8:ACAD:4::/64 [0/0]
via GigabitEthernet0/0/1, directly connected
L 2001:DB8:ACAD:4::1/128 [0/0]
via GigabitEthernet0/0/1, receive

O 2001:DB8:ACAD:5::/64 [110/2]
via FE80::B6A8:B9FF:FE47:92C0, GigabitEthernet0/0/1
O 2001:DB8:ACAD:6::/64 [110/3]
via FE80::B6A8:B9FF:FE47:92C0, GigabitEthernet0/0/1
OI 2001:DB8:ACAD:7::/64 [110/4]
via FE80::B6A8:B9FF:FE47:92C0, GigabitEthernet0/0/1
OI 2001:DB8:ACAD:8::/64 [110/5]
via FE80::B6A8:B9FF:FE47:92C0, GigabitEthernet0/0/1
L FF00::/8 [0/0]
via Null0, receive

R7#show ip ospf neighbor detail

Neighbor 8.8.8.8, interface address 192.168.4.2

In the area 0 via interface GigabitEthernet0/0/1

Neighbor priority is 1, State is FULL, 6 state changes

DR is 192.168.4.2 BDR is 192.168.4.1

Options is 0x12 in Hello (E-bit, L-bit)

Options is 0x52 in DBD (E-bit, L-bit, O-bit)

LLS Options is 0x1 (LR)

Dead timer due in 00:00:36

Neighbor is up for 00:54:10

Index 1/2/2, retransmission queue length 0, number of retransmission 0

First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0

Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 2.2.2.2, interface address 192.168.3.1

In the area 0 via interface GigabitEthernet0/0/0

Neighbor priority is 1, State is FULL, 6 state changes

DR is 192.168.3.1 BDR is 192.168.3.2

Options is 0x12 in Hello (E-bit, L-bit)

Options is 0x52 in DBD (E-bit, L-bit, O-bit)

LLS Options is 0x1 (LR)

Dead timer due in 00:00:32

Neighbor is up for 00:56:53

Index 1/1/1, retransmission queue length 0, number of retransmission 0

First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0

Last retransmission scan time is 0 msec, maximum is 0 msec

R7#show ipv6 ospf neighbor detail

OSPFv3 Router with ID (7.7.7.7) (Process ID 1)

Neighbor 8.8.8.8

In the area 0 via interface GigabitEthernet0/0/1

Neighbor: interface-id 6, link-local address FE80::B6A8:B9FF:FE47:92C0

Neighbor priority is 1, State is FULL, 6 state changes
DR is 8.8.8.8 BDR is 7.7.7.7
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:36
Neighbor is up for 00:37:45
Index 1/1/1, retransmission queue length 0, number of retransmission 1
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 2, maximum is 2
Last retransmission scan time is 1 msec, maximum is 1 msec

Neighbor 2.2.2.2

In the area 0 via interface GigabitEthernet0/0/0
Neighbor: interface-id 38, link-local address FE80::B28B:D0FF:FED6:C56
Neighbor priority is 1, State is FULL, 6 state changes
DR is 7.7.7.7 BDR is 2.2.2.2
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:33
Neighbor is up for 00:37:51
Index 1/2/2, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

R7#show ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi0/0/1	1	0	192.168.4.1/30	1	BDR	1/1	
Gi0/0/0	1	0	192.168.3.2/30	1	BDR	1/1	

Router 8

R8#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O IA 192.168.1.0/24 [110/4] via 192.168.4.1, 00:47:27, GigabitEthernet0/0/0

192.168.2.0/30 is subnetted, 1 subnets
O IA 192.168.2.0 [110/3] via 192.168.4.1, 00:47:27, GigabitEthernet0/0/0
192.168.3.0/30 is subnetted, 1 subnets
O 192.168.3.0 [110/2] via 192.168.4.1, 00:47:27, GigabitEthernet0/0/0
192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.4.0/30 is directly connected, GigabitEthernet0/0/0
L 192.168.4.2/32 is directly connected, GigabitEthernet0/0/0
192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.5.0/30 is directly connected, GigabitEthernet0/0/1
L 192.168.5.1/32 is directly connected, GigabitEthernet0/0/1
192.168.6.0/30 is subnetted, 1 subnets
O 192.168.6.0 [110/2] via 192.168.5.2, 00:44:40, GigabitEthernet0/0/1
192.168.7.0/30 is subnetted, 1 subnets
O IA 192.168.7.0 [110/3] via 192.168.5.2, 00:42:56, GigabitEthernet0/0/1
O IA 192.168.8.0/24 [110/4] via 192.168.5.2, 00:42:21, GigabitEthernet0/0/1

R8#show ipv6 route

IPv6 Routing Table - default - 11 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, H - NHRP, I1 - ISIS L1

I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP

EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination

NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1

OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2

a - Application

OI 2001:DB8:ACAD:1::/64 [110/4]
via FE80::B6A8:B9FF:FE01:B511, GigabitEthernet0/0/0
OI 2001:DB8:ACAD:2::/64 [110/3]
via FE80::B6A8:B9FF:FE01:B511, GigabitEthernet0/0/0
O 2001:DB8:ACAD:3::/64 [110/2]
via FE80::B6A8:B9FF:FE01:B511, GigabitEthernet0/0/0
C 2001:DB8:ACAD:4::/64 [0/0]
via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:4::2/128 [0/0]
via GigabitEthernet0/0/0, receive
C 2001:DB8:ACAD:5::/64 [0/0]
via GigabitEthernet0/0/1, directly connected
L 2001:DB8:ACAD:5::1/128 [0/0]
via GigabitEthernet0/0/1, receive
O 2001:DB8:ACAD:6::/64 [110/2]

R8#show ip ospf neighbor detail

Neighbor 9.9.9.9, interface address 192.168.5.2

In the area 0 via interface GigabitEthernet0/0/1

Neighbor priority is 1, State is FULL, 6 state changes

DR is 192.168.5.1 BDR is 192.168.5.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:33
Neighbor is up for 00:47:14
Index 1/2/2, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 7.7.7.7, interface address 192.168.4.1
In the area 0 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.4.2 BDR is 192.168.4.1
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:32
Neighbor is up for 00:48:49
Index 1/1/1, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
R8#show ipv6 ospf neighbor detail

OSPFv3 Router with ID (8.8.8.8) (Process ID 1)

Neighbor 9.9.9.9
In the area 0 via interface GigabitEthernet0/0/1
Neighbor: interface-id 6, link-local address FE80::B6A8:B9FF:FE01:AE50
Neighbor priority is 1, State is FULL, 6 state changes
DR is 9.9.9.9 BDR is 8.8.8.8
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:32
Neighbor is up for 00:34:56
Index 1/1/1, retransmission queue length 0, number of retransmission 1
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 1
Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 7.7.7.7
In the area 0 via interface GigabitEthernet0/0/0
Neighbor: interface-id 7, link-local address FE80::B6A8:B9FF:FE01:B511

Neighbor priority is 1, State is FULL, 6 state changes
 DR is 8.8.8.8 BDR is 7.7.7.7
 Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
 Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
 Dead timer due in 00:00:30
 Neighbor is up for 00:32:24
 Index 1/2/2, retransmission queue length 0, number of retransmission 0
 First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
 Last retransmission scan length is 0, maximum is 0
 Last retransmission scan time is 0 msec, maximum is 0 msec

R8#show ip ospf int brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi0/0/1	1	0	192.168.5.1/30	1	DR	1/1	
Gi0/0/0	1	0	192.168.4.2/30	1	DR	1/1	

Router 9

R9#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```

O IA 192.168.1.0/24 [110/5] via 192.168.5.1, 00:42:32, GigabitEthernet0/0/0
    192.168.2.0/30 is subnetted, 1 subnets
O IA 192.168.2.0 [110/4] via 192.168.5.1, 00:42:32, GigabitEthernet0/0/0
    192.168.3.0/30 is subnetted, 1 subnets
O 192.168.3.0 [110/3] via 192.168.5.1, 00:42:32, GigabitEthernet0/0/0
    192.168.4.0/30 is subnetted, 1 subnets
O 192.168.4.0 [110/2] via 192.168.5.1, 00:42:32, GigabitEthernet0/0/0
    192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.5.0/30 is directly connected, GigabitEthernet0/0/0
L 192.168.5.2/32 is directly connected, GigabitEthernet0/0/0
    192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.6.0/30 is directly connected, GigabitEthernet0/0/1
L 192.168.6.1/32 is directly connected, GigabitEthernet0/0/1
    192.168.7.0/30 is subnetted, 1 subnets
O IA 192.168.7.0 [110/2] via 192.168.6.2, 00:39:00, GigabitEthernet0/0/1
  
```

O IA 192.168.8.0/24 [110/3] via 192.168.6.2, 00:38:25, GigabitEthernet0/0/1

R9#show ipv6 route

IPv6 Routing Table - default - 11 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, H - NHRP, I1 - ISIS L1

I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP

EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination

NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1

OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2

a - Application

OI 2001:DB8:ACAD:1::/64 [110/5]

via FE80::B6A8:B9FF:FE47:92C1, GigabitEthernet0/0/0

OI 2001:DB8:ACAD:2::/64 [110/4]

via FE80::B6A8:B9FF:FE47:92C1, GigabitEthernet0/0/0

O 2001:DB8:ACAD:3::/64 [110/3]

via FE80::B6A8:B9FF:FE47:92C1, GigabitEthernet0/0/0

O 2001:DB8:ACAD:4::/64 [110/2]

via FE80::B6A8:B9FF:FE47:92C1, GigabitEthernet0/0/0

C 2001:DB8:ACAD:5::/64 [0/0]

via GigabitEthernet0/0/0, directly connected

L 2001:DB8:ACAD:5::2/128 [0/0]

via GigabitEthernet0/0/0, receive

C 2001:DB8:ACAD:6::/64 [0/0]

via GigabitEthernet0/0/1, directly connected

L 2001:DB8:ACAD:6::1/128 [0/0]

via GigabitEthernet0/0/1, receive

OI 2001:DB8:ACAD:7::/64 [110/2]

via FE80::521C:B0FF:FE2D:7100, GigabitEthernet0/0/1

OI 2001:DB8:ACAD:8::/64 [110/3]

via FE80::521C:B0FF:FE2D:7100, GigabitEthernet0/0/1

L FF00::/8 [0/0]

via Null0, receive

R9#show ip ospf neighbor detail

Neighbor 10.10.10.10, interface address 192.168.6.2

In the area 0 via interface GigabitEthernet0/0/1

Neighbor priority is 1, State is FULL, 6 state changes

DR is 192.168.6.1 BDR is 192.168.6.2

Options is 0x12 in Hello (E-bit, L-bit)

Options is 0x52 in DBD (E-bit, L-bit, O-bit)

LLS Options is 0x1 (LR)

Dead timer due in 00:00:39

Neighbor is up for 00:40:09

Index 1/2/2, retransmission queue length 0, number of retransmission 0

First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 8.8.8.8, interface address 192.168.5.1
In the area 0 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.5.1 BDR is 192.168.5.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:33
Neighbor is up for 00:43:02
Index 1/1/1, retransmission queue length 0, number of retransmission 1
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 1
Last retransmission scan time is 0 msec, maximum is 0 msec
R9#show ipv6 ospf neighbor detail

OSPFv3 Router with ID (9.9.9.9) (Process ID 1)

Neighbor 10.10.10.10
In the area 0 via interface GigabitEthernet0/0/1
Neighbor: interface-id 6, link-local address FE80::521C:B0FF:FE2D:7100
Neighbor priority is 1, State is FULL, 6 state changes
DR is 10.10.10.10 BDR is 9.9.9.9
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:32
Neighbor is up for 00:33:34
Index 1/1/1, retransmission queue length 0, number of retransmission 1
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 1
Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 8.8.8.8
In the area 0 via interface GigabitEthernet0/0/0
Neighbor: interface-id 7, link-local address FE80::B6A8:B9FF:FE47:92C1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 9.9.9.9 BDR is 8.8.8.8
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:38
Neighbor is up for 00:30:54
Index 1/2/2, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0

Last retransmission scan time is 0 msec, maximum is 0 msec

R9#show ip ospf int brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi0/0/1	1	0	192.168.6.1/30	1	DR	1/1	
Gi0/0/0	1	0	192.168.5.2/30	1	BDR	1/1	

Router 10

R10#show run

Building configuration...

Current configuration : 1628 bytes

!

! Last configuration change at 19:00:21 UTC Wed Sep 24 2025

!

version 15.5

service timestamps debug datetime msec

service timestamps log datetime msec

no platform punt-keepalive disable-kernel-core

!

hostname R10

!

boot-start-marker

boot-end-marker

!

!

vrf definition Mgmt-intf

!

address-family ipv4

exit-address-family

!

address-family ipv6

exit-address-family

!

!

no aaa new-model

!

!

!

!

!

!

!

```
!  
!  
!  
!  
  
!  
!  
!  
ipv6 unicast-routing  
!  
!  
!  
!  
!  
!  
!  
subscriber templating  
vtp domain cisco  
vtp mode transparent  
multilink bundle-name authenticated  
!  
!  
!  
!  
license udi pid ISR4321/K9 sn FDO21491FHX  
!  
spanning-tree extend system-id  
!  
!  
redundancy  
mode none  
!  
!  
vlan internal allocation policy ascending  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0/0  
ip address 192.168.6.2 255.255.255.252  
ip ospf 1 area 0
```

```
negotiation auto
ipv6 address 2001:DB8:ACAD:6::2/64
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/0/1
ip address 192.168.7.1 255.255.255.252
ip ospf 1 area 51
negotiation auto
ipv6 address 2001:DB8:ACAD:7::1/64
ipv6 ospf 1 area 51
!
interface GigabitEthernet0/1/0
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet0/1/1
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
router-id 10.10.10.10
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
ipv6 router ospf 1
router-id 10.10.10.10
!
!
```

```

!
!
control-plane
!
!
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
!
!
end

```

R10#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```

O IA 192.168.1.0/24 [110/6] via 192.168.6.1, 00:25:57, GigabitEthernet0/0/0
    192.168.2.0/30 is subnetted, 1 subnets
O IA 192.168.2.0 [110/5] via 192.168.6.1, 00:25:57, GigabitEthernet0/0/0
    192.168.3.0/30 is subnetted, 1 subnets
O 192.168.3.0 [110/4] via 192.168.6.1, 00:25:57, GigabitEthernet0/0/0
    192.168.4.0/30 is subnetted, 1 subnets
O 192.168.4.0 [110/3] via 192.168.6.1, 00:25:57, GigabitEthernet0/0/0
    192.168.5.0/30 is subnetted, 1 subnets
O 192.168.5.0 [110/2] via 192.168.6.1, 00:25:57, GigabitEthernet0/0/0
    192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks

```

R10#show ipv6 route

IPv6 Routing Table - default - 11 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, H - NHRP, I1 - ISIS L1

I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP

EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination

NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1

OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
a - Application

OI 2001:DB8:ACAD:1::/64 [110/6]
via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
OI 2001:DB8:ACAD:2::/64 [110/5]
via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
O 2001:DB8:ACAD:3::/64 [110/4]
via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
O 2001:DB8:ACAD:4::/64 [110/3]
via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
O 2001:DB8:ACAD:5::/64 [110/2]
via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
C 2001:DB8:ACAD:6::/64 [0/0]
via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:6::2/128 [0/0]
via GigabitEthernet0/0/0, receive
C 2001:DB8:ACAD:7::/64 [0/0]

R10#show ip ospf neighbor detail

Neighbor 9.9.9.9, interface address 192.168.6.1

In the area 0 via interface GigabitEthernet0/0/0

Neighbor priority is 1, State is FULL, 6 state changes

DR is 192.168.6.1 BDR is 192.168.6.2

Options is 0x12 in Hello (E-bit, L-bit)

Options is 0x52 in DBD (E-bit, L-bit, O-bit)

LLS Options is 0x1 (LR)

Dead timer due in 00:00:30

Neighbor is up for 00:27:23

Index 1/1/1, retransmission queue length 0, number of retransmission 0

First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0

Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 11.11.11.11, interface address 192.168.7.2

In the area 1 via interface GigabitEthernet0/0/1

Neighbor priority is 1, State is FULL, 6 state changes

DR is 192.168.7.2 BDR is 192.168.7.1

Options is 0x12 in Hello (E-bit, L-bit)

Options is 0x52 in DBD (E-bit, L-bit, O-bit)

LLS Options is 0x1 (LR)

Dead timer due in 00:00:36

Neighbor is up for 00:26:41

Index 1/1/2, retransmission queue length 0, number of retransmission 0

First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0

Last retransmission scan time is 0 msec, maximum is 0 msec
R10#show ipv6 ospf neighbor detail

OSPFv3 Router with ID (10.10.10.10) (Process ID 1)

Neighbor 9.9.9.9

In the area 0 via interface GigabitEthernet0/0/0
Neighbor: interface-id 7, link-local address FE80::B6A8:B9FF:FE01:AE51
Neighbor priority is 1, State is FULL, 6 state changes
DR is 10.10.10.10 BDR is 9.9.9.9
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:38
Neighbor is up for 00:20:32
Index 1/1/1, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 11.11.11.11

In the area 1 via interface GigabitEthernet0/0/1
Neighbor: interface-id 6, link-local address FE80::6EDD:30FF:FEB6:6030
Neighbor priority is 1, State is FULL, 6 state changes
DR is 10.10.10.10 BDR is 11.11.11.11
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:34
Neighbor is up for 00:19:23
Index 1/1/2, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

R10#show ip ospf int brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi0/0/0	1	0	192.168.6.2/30	1	BDR	1/1	
Gi0/0/1	1	1	192.168.7.1/30	1	BDR	1/1	

R10#show ip ospf database

OSPF Router with ID (10.10.10.10) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
2.2.2.2	2.2.2.2	215	0x8000003B	0x00F616	1
7.7.7.7	7.7.7.7	1954	0x80000006	0x00E447	2
8.8.8.8	8.8.8.8	51	0x80000005	0x00EA36	2
9.9.9.9	9.9.9.9	1720	0x80000005	0x00CC49	2

10.10.10.10 10.10.10.10 1687 0x80000003 0x0055A8 1

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.3.1	2.2.2.2	215	0x80000002	0x00168C
192.168.4.2	8.8.8.8	51	0x80000002	0x001957
192.168.5.1	8.8.8.8	51	0x80000002	0x007CEB
192.168.6.1	9.9.9.9	1720	0x80000001	0x00A9B2

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.1.0	2.2.2.2	457	0x80000002	0x00A225
192.168.2.0	2.2.2.2	457	0x80000002	0x007B4F
192.168.7.0	10.10.10.10	1677	0x80000001	0x005551
192.168.8.0	10.10.10.10	1642	0x80000001	0x00663B

Router Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
10.10.10.10	10.10.10.10	1647	0x80000003	0x006794	1
11.11.11.11	11.11.11.11	1647	0x80000005	0x00D39B	2

Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.7.2	11.11.11.11	1647	0x80000001	0x009CAD

Summary Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.1.0	10.10.10.10	1687	0x80000001	0x00DBC8
192.168.2.0	10.10.10.10	1687	0x80000001	0x00B4F2
192.168.3.0	10.10.10.10	1687	0x80000001	0x009F08
192.168.4.0	10.10.10.10	1687	0x80000001	0x008A1D
192.168.5.0	10.10.10.10	1687	0x80000001	0x007532
192.168.6.0	10.10.10.10	1687	0x80000001	0x006047

Router 11

R11#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O IA 192.168.1.0/24 [110/7] via 192.168.7.1, 00:48:39, GigabitEthernet0/0/0
 192.168.2.0/30 is subnetted, 1 subnets
 O IA 192.168.2.0 [110/6] via 192.168.7.1, 00:48:39, GigabitEthernet0/0/0
 192.168.3.0/30 is subnetted, 1 subnets
 O IA 192.168.3.0 [110/5] via 192.168.7.1, 00:48:39, GigabitEthernet0/0/0
 192.168.4.0/30 is subnetted, 1 subnets
 O IA 192.168.4.0 [110/4] via 192.168.7.1, 00:48:39, GigabitEthernet0/0/0
 192.168.5.0/30 is subnetted, 1 subnets
 O IA 192.168.5.0 [110/3] via 192.168.7.1, 00:48:39, GigabitEthernet0/0/0
 192.168.6.0/30 is subnetted, 1 subnets
 O IA 192.168.6.0 [110/2] via 192.168.7.1, 00:48:39, GigabitEthernet0/0/0
 192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks
 C 192.168.7.0/30 is directly connected, GigabitEthernet0/0/0
 L 192.168.7.2/32 is directly connected, GigabitEthernet0/0/0
 192.168.8.0/24 is variably subnetted, 2 subnets, 2 masks
 C 192.168.8.0/24 is directly connected, GigabitEthernet0/0/1
 L 192.168.8.1/32 is directly connected, GigabitEthernet0/0/1

R11# show ipv6 route

IPv6 Routing Table - default - 11 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

OI 2001:DB8:ACAD:1::/64 [110/7]
 via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0
 OI 2001:DB8:ACAD:2::/64 [110/6]
 via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0
 OI 2001:DB8:ACAD:3::/64 [110/5]
 via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0
 OI 2001:DB8:ACAD:4::/64 [110/4]
 via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0
 OI 2001:DB8:ACAD:5::/64 [110/3]
 via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0
 OI 2001:DB8:ACAD:6::/64 [110/2]
 via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0
 C 2001:DB8:ACAD:7::/64 [0/0]
 via GigabitEthernet0/0/0, directly connected
 L 2001:DB8:ACAD:7::2/128 [0/0]
 via GigabitEthernet0/0/0, receive
 C 2001:DB8:ACAD:8::/64 [0/0]
 via GigabitEthernet0/0/1, directly connected
 L 2001:DB8:ACAD:8::1/128 [0/0]

via GigabitEthernet0/0/1, receive
L FF00::/8 [0/0]
via Null0, receive

R11#show ip ospf neigh detail

Neighbor 10.10.10.10, interface address 192.168.7.1
In the area 1 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.7.2 BDR is 192.168.7.1
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:34
Neighbor is up for 00:50:31
Index 1/1/1, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

R11#show ipv6 ospf neigh de

OSPFv3 Router with ID (11.11.11.11) (Process ID 1)

Neighbor 10.10.10.10
In the area 1 via interface GigabitEthernet0/0/0
Neighbor: interface-id 7, link-local address FE80::521C:B0FF:FE2D:7101
Neighbor priority is 1, State is FULL, 6 state changes
DR is 10.10.10.10 BDR is 11.11.11.11
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:38
Neighbor is up for 00:43:46
Index 1/1/1, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

R11#show ip ospf int brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi0/0/0	1	1	192.168.7.2/30	1	DR	1/1	
Gi0/0/1	1	1	192.168.8.1/24	1	DR	0/0	

Problems

During this lab we faced some layer 1 problems with the wiring between the routers as we thought the routers were cabled differently than expected, but we were able to solve this quickly by fixing the wiring to match the topology.

Conclusion

This lab showed the fundamentals of enterprise networking and how layer 3 devices communicate to each other to access servers and other end devices. The lab was relatively simple and fun to configure.